

On the Effectiveness of Virtual IMU Data for Eating Detection with Wrist Sensors

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ABSTRACT

The successful training of human activity recognition (HAR) systems typically heavily depends on the availability of sufficient amounts of labeled sensor data. Unfortunately, obtaining large-scale labeled datasets is usually expensive and often limited by practical and / or privacy reasons. Recently, IMUTube was introduced to tackle this data scarcity problem through generating weakly-labeled, virtual IMU data from unconstrained video repositories, such as YouTube. IMUTube was demonstrated to be very effective at classifying locomotion or gym exercises that involve large movements of body parts. Yet, many important daily activities, such as eating, do not exhibit such substantial body (part) movements but are rather based on more subtle, fine-grained motions. This work explores the utility of IMUTube for such subtle motion activities with specific, exemplary application to eating detection. We found that—surprisingly—IMUTube is also very effective for this challenging HAR domain. Our experiment demonstrates that eating recognition systems benefit from virtual IMU data extracted from video datasets with significant improvements of recognition accuracy (increases of 8.4% and 5.9% F1-score absolute for both curated and in-the-wild video datasets, respectively relative to 71.5% F1-score of the baseline), which is encouraging for the broader use of systems like IMUTube.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous computing**; • **Computing methodologies** → *Activity recognition and understanding*.

KEYWORDS

Virtual IMU data, Daily Activity Recognition, Eating recognition

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1 INTRODUCTION

The effectiveness of supervised learning methods for deriving human activity recognition systems (HAR) for wearables depends heavily on the availability of curated, i.e., annotated datasets [5]. One major issue with current machine learning solutions in the field is the paucity of specialized labeled datasets. Annotating sensor data in HAR is expensive, often privacy invasive, and often prone to errors or other practical limitations [8]. Recently, systems like IMUTube [9] have been introduced that tackle the aforementioned problem by generating virtual IMU data from videos to increase the size of training datasets that can be used for model training. However, the main focus of systems like IMUTube so far was on coarse motion activities, i.e., those that involve significant movements of body parts like locomotion or gym exercises. Yet, many important activities in daily life are more subtle, exhibiting much lesser motion differences between and more variance within activities, e.g., eating, holding (and talking on) a phone, or washing dishes, etc. [10]

In this work, we take a step into exploring the versatility of IMUTube in generating virtual IMU data on *subtle motion* activities. We consider eating activity as one of the primary examples of fine motion activity [1–3, 10]. It involves repeated hand movement to the mouth, which varies vastly depending on the eating habits and food consumed. There are also lots of similar movements in daily living that can create false positives for such detection methods. Eating detection is a mature field and many studies have been conducted in which a range of wrist based sensing and analysis systems have been utilized [2–4].

We build on this related work and evaluate to what extent the aforementioned eating detection systems can potentially benefit from a system like IMUTube. Much to our surprise, we found that virtual IMU data from IMUTube can very effectively capture eating motions from videos related to eating. This is important because IMUTube was designed with coarse, i.e., significant body motion in mind, which is in contrast to the subtle hand to mouth movements in eating. In our experiments, we could improve eating detection performance by 8.4% absolute when using curated video data, and by 5.9% absolute (both F1-scores) when using unconstrained video data as they were retrieved through keyword-based search on YouTube. These results encourage the HAR community to further investigate the use of cross-modality transfer approaches such as IMUTube for various complex and subtle human activities in the real world.

2 EXPERIMENT

To evaluate the effectiveness of IMUTube on eating recognition tasks, we followed the approach used in the original IMUTube experiments [9]. We extracted virtual IMU data for wrist sensors from video datasets, which also includes calibration with some

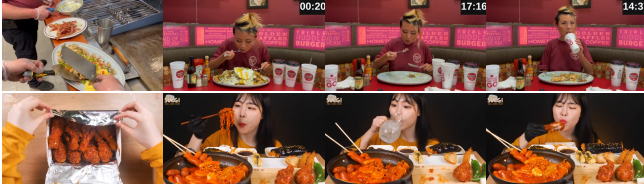


Figure 1: In-wild YouTube video instances

real IMU data, and then trained a Random Forest model with both calibrated virtual IMU data and (the small amounts of) real IMU data. The trained model was then tested on in-the wild real IMU data to assess the ecological validity of the trained model.

For calibrating the virtual IMU data we used the Lab-20 dataset [10] where data recording was more restricted. This choice was motivated by previous work [1–3, 10], which focus to develop eating moment recognition solution in controlled setting that can work well in-wild. The Lab-20 dataset was collected from 21 participants covering both eating and non-eating activities for approximately 31 minutes period on average. Eating activities include the use of various utensils, such as fork, knife, spoon, chopsticks, or plain hands. Non-eating activities include daily activities that involves hand and arm movements that are similar to eating activities, such as brushing teeth, combing hair, and holding a phone. For in-the-wild experiments, we used the Wild-7 dataset [10], which was collected from seven participants throughout the day using wrist sensors and wearable cameras. Out of a total of 31 hours and 28 minutes of data, 2 hours and 8 minutes were labeled as eating activities. Similar to previous work [10], we applied a DBSCAN [6] based analysis approach to determine eating moments.

We tested on two sets of video data: *i*) curated; and *ii*) in-the-wild. For the former we used videos from Kinetics-400 [7], where all videos were manually trimmed down to clips of 10 seconds to contain specific activity classes. In our experiments, we used 417 video clips for the eating class, which were labeled as one of 10 eating-related classes (eating: burger, cake, carrots, chips, doughnuts, hotdog, ice cream, nachos, spaghetti, watermelon). The extracted virtual IMU data contained approximately 6% more eating samples compared to in-lab dataset. For in-the-wild experiments, we downloaded two YouTube videos queried with ‘eating’ keyword, as shown in Figure 1. The first video is 22 minutes long, whereas the second one has a duration of 17 minutes. Both videos combined, the extracted virtual IMU data resulted in approximately 11% more eating samples compared to the in-lab dataset.

For model training and evaluation, we sub-sampled the real IMU data to 25 Hz to match the sampling rate of the virtual IMU dataset extracted from videos. Following previous eating studies [10], we applied sliding window segmentation with 6 second window size and 50% overlap between consecutive analysis frames. For each frame we then extracted mean, variance, skewness, kurtosis and root mean square features, which is then fed into model training (Random Forest). We report binary F1-scores from the average of three runs on our experiment.

3 RESULTS

Our baseline model has been determined following the protocol from previous work [10], which trained models only with in-lab

Table 1: F1-scores on Wild-7 dataset for eating moment recognition task. Note that Real data means Lab-20 dataset. We want to highlight that due to standardization, we are getting a lower F1-score compared to [10].

Training data source	F1-score for eating class
Real data only [10]	71.5
Real + Kinetics-400	79.94
Real + YouTube	77.45

data. As mentioned before, our model is trained with virtual IMU data either from curated or in-the-wild video datasets along with (small amounts of) real IMU data. Then, the models are evaluated on in-the-wild test dataset. Results are shown in Table 1: the baseline model achieved 71.5% F1-score, and our model trained with the curated virtual IMU dataset improved that to 79.94%. The in-the-wild-scenario increased the recognition performance by 5.9% to 77.45%.

4 DISCUSSION AND CONCLUSION

In this work, we studied the utility of IMUTube for subtle motions prevalent in everyday activity with an exemplary focus on eating. Our experiment demonstrated surprising effectiveness of IMUTube in analysis domains where there are only subtle, fine-grained movements – in contrast to the more coarse movements that IMUTube was designed to tackle originally. It also showed that the quality of video can affect the resulting quality of the extracted virtual IMU data as the model trained with curated dataset outperformed the model trained with in-the-wild data. To conclude: this work shows new opportunities for using virtual IMU data even in HAR domains where only subtle movements of body (parts) are to be observed.

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